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The Functionality Appreciation Scale: An examination of its psychometric properties in a French-Canadian adult sample☆

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ABSTRACT

The 7-item Functionality Appreciation Scale (FAS) is a widely used instrument to assess functionality appreciation, a central facet of positive body image. Here, we examined the psychometric properties of a novel French translation of the FAS. French-speaking Canadian adults were asked to complete the FAS and validated measures of positive body image, psychological well-being, and eating patterns. In a first split-half subsample ($n = 385$), exploratory factor analysis supported extraction of a unidimensional model with all 7 items. In a second split-half subsample ($n = 385$), confirmatory factor analytic findings supported the unidimensional model of the FAS. A partial factors-only model indicated that, in both subsamples, the French FAS had differential item functioning based on respondent age and body mass index (BMI), but not on gender identity. In both subsamples, higher respondent age was associated with greater functionality appreciation, whereas higher BMI was associated with lower functionality appreciation. Additional results supported convergent and concurrent validity of the French FAS, with greater functionality appreciation being associated with greater body appreciation and psychological well-being, as well as lower symptoms of maladaptive eating patterns. The present results indicate that the French FAS has strong psychometric properties in French-speaking Canadian adults.

1. Introduction

In a seminal article published almost a decade ago, Tylka and Wood-Barcalow (2015a) defined *positive body image* as a multifaceted, stable, yet malleable construct (i.e., including both trait and state qualities), encompassing protective capacities for physical and psychological health and well-being. This multifacetedness is reflected in the wide range of facets of positive body image that have been identified, such as body appreciation (Tylka & Wood-Barcalow, 2015b) and functionality appreciation (Alleva et al., 2017). *Functionality appreciation* – an important facet of positive body image (Swami et al., 2020) – is defined as a "appreciating, respecting, and honouring the body for what it is capable of doing, extending beyond mere awareness of body functionality" (Alleva et al., 2017, p. 29). That is, although functionality appreciation is related to body functionality (i.e., what the body is capable of doing; Alleva & Tylka, 2021), it is only theorised as a body image construct when it includes elements of gratitude, appreciation, or respect for what the body can do (Alleva & Martijn, 2019; Alleva & Tylka, 2021).

To assess the construct of functionality appreciation, Alleva et al. (2017) developed the Functionality Appreciation Scale (FAS). To do so, these authors initially constructed a pool of 26 items that reflected body functionality holistically (i.e., referring to diverse functional domains, such as one's physical capabilities, sensory experiences, and creative endeavours) and inclusively (i.e., capturing the overall appreciation of the body's ability to function the best it can). In an online sample of US-based adults, findings from exploratory factor analysis (EFA) supported the exclusion of 19 items (e.g., those that overlapped in terms of

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content or that had poor item-factor loadings). A follow-up EFA and confirmatory factor analysis (CFA) with new samples from the United States supported extraction of a unidimensional model of FAS consisting of 7 items. This model was also reported to achieve strong invariance across gender identity, test-retest reliability up to three weeks, and adequate construct validity (convergent, criterion-related, divergent, and predictive validity; Alleva et al., 2017).

Since its development, the validity of the unidimensional model of the FAS has been supported in a diverse range of communities (for a review, see Alleva, 2025). For example, in the United States, the unidimensionality of the FAS has been supported in adults of minoritised sexual identity (Soulliard & Vander Wal, 2021, 2022) and in Hispanic/Latina women (Pacheco et al., 2024). Additionally, the unidimensionality of the FAS has also been supported in English-speaking adolescents from the United Kingdom (Todd et al., 2019) and an international sample of adults (Linardon et al., 2020). Beyond English-speaking samples, we are aware of 16 translations of the FAS that have been validated in distinct national groups (for a summary, see Table 1). In all of these cases, the unidimensional model of the FAS has been upheld, typically through the use of a combination of EFA and CFA (e.g., Cerea et al., 2021; Mebarak et al., 2023; Swami et al., 2021). Composite reliability coefficients are typically very good (>.85) when assessed using Cronbach's α and latterly McDonald's (1970) ω.

Beyond issues of factor structure, many validation studies have also assessed measurement invariance of the FAS (i.e., the extent to which the FAS measures the same latent construct) across gender identity (e.g., Anastasiades et al., 2023; Riechers & Warschburger, 2025). Typically, these studies have established at least strong (or scalar) invariance, which in turn has allowed for confident comparisons of functionality appreciation across women and men. Comparative analyses have generally indicated that there are no significant gender differences in functionality appreciation, which is consistent with the findings of one recent meta-analysis, which reported a non-significant, weak pooled effect size ($d = 0.08$; Linardon et al., 2023). Additionally, one study has reported the FAS achieved strong invariance across age groups, where age was categorised in terms of middle school students, high school students, young adults, and older adults (He et al., 2023). The meta-analysis by Linardon et al. (2023) also indicated no significant association between functionality appreciation and age ($r = .02$).

To date, however, most examinations of the invariance of the FAS have utilised multi-group CFA, which is suitable when groups are treated as categories (e.g., national groups, gender identities; Chen, 2007). However, where characteristics occur as continuous variables (e.g., age and body mass index) or when testing for measurement biases as a joint function of multiple variables concurrently, it is recommended that tests of differential item functioning (DIF) are conducted instead (Penfield & Lam, 2000). At present, we are aware of only one study that has applied this analytic strategy. In a study utilising data from Colombian and Spanish respondents, Mebarak et al. (2024) reported that the FAS lacked DIF across age and body mass index (BMI), suggesting that the FAS is measuring the same latent construct across participant age and BMI in these national groups.

Finally, consistent with the findings of the parent study (Alleva et al., 2017), studies in diverse national contexts have supported the construct and nomological validity of the FAS. Perhaps the most widely reported finding is of significant positive associations with other facets of positive body image – mainly body appreciation – and negative associations with symptoms of disordered eating, which present evidence of the instrument's convergent validity (e.g., Mebarak et al., 2023; Swami et al., 2019). FAS scores are also associated with indices of psychological well-being (e.g., self-esteem, life satisfaction, e.g., Alleva et al., 2023), providing concurrent validity support. These patterns of findings are corroborated by a meta-analysis of the correlates of functionality appreciation, which found that the construct was consistently associated with fewer body image problems, lower levels of disordered eating, and more positive mental health (Linardon et al., 2023). In short, the psychometric properties of the FAS appear to be robust across diverse populations.

1.1. *The present study*

The main objective of the present study was to add to the growing test adaptation literature on the FAS by examining the psychometric properties of a novel French translation in a sample of French-Canadian adults. Although a French translation of the FAS has recently been prepared and validated in adults from France (Arnaud et al., 2025), the translation method utilised by the authors used only forward-translation; that is, there was an absence of back-translation and additional translational steps (for discussions on best practices for instrument translation, see Swami & Barron, 2019; Swami et al., 2021).

Table 1
Summary of test adaptation studies of the Functionality Appreciation Scale with reference to languages and National groups.

Language	National Group	Sample Type	Analytic Strategy for Factorial Validity	Measurement Invariance	Reference
Arabic	Lebanon	Adults	EFA, CFA	Gender identity (strong)	Swami et al. (2022)
Chinese	China	Mixed	EFA, CFA	Gender identity (strong), age groups (strong)	He et al. (2022)
Dutch	The Netherlands	Adults	EFA only	Gender identity (strict)	Alleva et al. (2023)
Dutch	The Netherlands	Adults	EFA, CFA	Sample type (community vs. eating disorder patients; weak)	Rekkers et al. (2025)
English	United Kingdom	Adolescents	CFA only	Not assessed	Todd et al. (2019)
English	United States	Adults	EFA, CFA	Gender identity (strong)	Alleva et al. (2017)
Farsi	Iran	Adolescents	EFA, CFA	Gender identity (strong)	Sahlan et al. (2022)
French	France	Adults	EFA, CFA	Gender identity (strong)	Arnaud et al. (2025)
German	Germany	Adults	EFA, CFA	Gender identity (strict)	Riechers and Warschburger (2025)
Greek	Cyprus	Adults	EFA, CFA	Gender identity (strong)	Anastasiades et al. (2023)
Italian	Italy	Adults	EFA, CFA	Gender identity (strong)	Cerea et al. (2021)
Japanese	Japan	Adults	EFA, CFA	Gender identity (strong)	Namatame et al. (2022)
Lithuanian	Lithuania	Adults	EFA, CFA	Gender identity (strong)	Baceviciene et al. (2025)
Malay	Malaysia	Adults	EFA, CFA	Gender identity (strong)	Swami et al. (2019)
Polish	Poland	Adults	Not assessed	Not assessed	Yurtsever et al. (2022)
Portuguese	Brazil	Adults	CFA only	Not assessed	Faria et al. (2020)
Romanian	Romania	Adults	EFA, CFA	Gender identity (strong)	Swami et al. (2021)
Spanish	Colombia	Adults	EFA, CFA	Gender identity (strong)	Mebarak et al. (2023)
Spanish	Spain	Adults	EFA, CFA	Gender identity (strong)	Zamora et al. (2024)

Note. FA = exploratory factor analyses; CFA = confirmatory factor analysis.

Additionally, and importantly, Canadian French contains vocabulary differences from Metropolitan French that impact the wording of certain FAS items. This necessitates a novel translation of the FAS, alongside its validation for use in French-Canadian adults. Indeed, it should not be assumed that findings from French-speaking adults in France will necessarily transfer to French-speaking adults in Canada.

There are additional reasons why it is important to assess the psychometric properties of the FAS in French-Canadians adults. For example, life in Quebec – the only Canadian province with a French-speaking majority – is characterised by highly variable geography, land use, and weather (e.g., high snowfall and cold temperatures in winter), all of which may detrimentally impact levels of physical activity and by extension functionality appreciation (e.g., Gagnon et al., 2022; Lebel et al., 2009). In fact, the Québécois administration has acknowledged that promoting physical activity is a priority (Ministère de l'Éducation du Loisir et du Sport du Québec, 2021). Although current policy around physical activity is primarily aimed at children and adolescents (on the assumption that recurrent physical activity in these age groups will transfer into adulthood; Hebinck et al., 2023), it is also important to consider barriers and motivators to physical activity in adulthood (Le Bodo et al., 2017). A better understanding of functionality appreciation and its correlates in this regional context, therefore, remains important and would be facilitated by the availability of a validated French-Canadian version of the FAS.

Thus, we followed best-practice guidelines to first develop our novel French translation of the FAS (Swami & Barron, 2019) and then to assess its psychometric properties, including its factor structure, by adopting an EFA-to-CFA strategy (Swami & Barron, 2019; Swami, Maïano et al., 2023). More specifically, EFA allows for an examination of factor structure without the imposition of modelling constraints (i.e., it is a data-driven approach to factor retention), while CFA allows for a cross-validation of the model (presumably unidimensional) derived in the first-pass analytic step. Based on previous work (e.g., Alleva et al., 2017, 2023), we expected that scores on our French translation of the FAS would conform to a unidimensional structure with all seven items retained. Additionally, we examined invariance of the best factor solution across two independent samples and DIF based on participant gender identity (women vs. men), age, and BMI, as well as its invariance as function of split-half subsamples. Here, we expected that the FAS would be invariant across the subsamples and exhibit a lack of DIF as a function of respondent characteristics, which would be consistent with recent findings (Mebarak et al., 2024).

Additionally, we examined the convergent and concurrent validity of the French FAS. To do so, we considered associations between FAS scores and scores on additional variables that have been validated for use in French-speaking populations. More specifically, to assess convergent validity, we examined associations with a facet of positive body image (i.e., body appreciation) and eating patterns (i.e., dieting behaviour, orthorexia), with the expectation of moderate associations (positive with body appreciation, negative with dieting behaviour and orthorexia, respectively). We also examined concurrent validity of the FAS via associations with psychological well-being (symptoms of anxiety and depression, life satisfaction, self-esteem) and self-compassion. Here, we expected positive and small-to-moderate associations between functionality appreciation and life satisfaction, self-esteem, and self-compassion, and negative and small-to-moderate associations with anxiety and depression.

2. Method

2.1. Participants

A sample of 770 French-speaking Canadian adults, ranging in age from 17 to 90 years ($M = 50.87$, $SD = 15.87$), participated in the present study. Of the total sample, 409 (53.1%) identified as women, 352 (45.7%) as men, 5 (0.6%) as non-binary, and 4 (0.5%) as another gender identity. In terms of educational qualifications, 0.3% had completed elementary education, 14.2% had completed high school education, 15.5% had completed a diploma of vocational studies, 22.6% had completed a pre-university programme (CEGEP), 30.6% had an undergraduate degree, and 16.8% had a postgraduate degree. In total, 4.1% and 6.3% respectively reported currently having an eating disorder diagnosis or having received one in the past, while 0.7% of women reported being pregnant. Finally, participants ranged in self-reported BMI from 11.48 to 55.35 kg/m^2 ($M = 27.56$, $SD = 6.44$).

2.2. Measures

2.2.1. Participant information

Participants were asked to self-report demographics (i.e., age, gender identity, and educational qualifications) and anthropometric (weight and height) information. BMI (in kg/m^2) was calculated based on weight and height data. In addition, they were asked to report if they currently had a diagnosis of eating disorder or if they had received one in the past (0 = *no*; 1 = *yes*). Women were also asked if they were pregnant. Data on eating disorder diagnosis and pregnancy were requested for sample descriptive purposes only.

2.2.2. Functionality appreciation

To prepare a French translation of the FAS from English, we followed the recommendations of Swami and Barron (2019). More specifically, two independent translators (i.e., one professional translator and one author of the present study, who is bilingual) separately forward-translated the FAS (i.e., its instructions, response categories, and items). Next, a third independent translator (a member of the research team) examined the two French forward-translations. Minor discrepancies between the two translations were resolved and the translator produced a synthesised French version. Third, this synthetised French version was separately back-translated into English by two independent professional translators. Finally, all translators and all members of the research team examined, discussed, and resolved minor discrepancies between the forward- and back-translations. The final French version utilised in the present study is presented in the Supplementary Materials. We were not aware that Arnaud et al. (2025) had produced their own French translation at the time that we started this project. A side-by-side comparison of the two translations indicate that there are word-choice differences on three items (Items 2, 5, and 6).

2.2.3. Body appreciation

To assess body appreciation, we used the Body Appreciation Scale-2 (BAS-2; Tylka & Wood-Barcalow, 2015b; French translation: Kertechian & Swami, 2017). This 10-item instrument assesses acceptance of one's body, respect and care for one's body, and protection of one's body from unrealistic beauty standards (sample item: "I feel love for my body"). Participants rated items on a 5-point scale (1 = *never*, 5 = *always*), with higher scores reflecting greater body appreciation. In French-speaking Canadian adults (Swami et al., 2023), the BAS-2 has been shown to have a unidimensional factor structure, adequate composite reliability, and good construct validity. In the present study, the composite reliability coefficient (McDonald's ω) of scores on this scale was .958.

2.2.4. Dieting

The 13-item Dieting subscale of the Eating Attitudes Test-26 (EAT-26; Garner et al., 1982; French translation: Leichner et al., 1994) was used to measure dieting-related attitudes and behaviours characteristic of disordered eating. Participants rated items on a 6-point scale ranging from 1 (*never*) to 6 (*always*), with higher scores reflective of greater dieting attitudes and behaviours (sample item: "I am preoccupied with a desire to be thinner"). In French-speaking Canadian adults (Leichner et al., 1994), the EAT-26, including the Dieting subscale, has been shown to have adequate construct validity and composite reliability. In the present study, the composite reliability coefficient (McDonald's ω) of scores on this subscale was .928.

2.2.5. Orthorexia

We asked participants to complete the Teruel Orthorexia Scale (TOS; Barrada & Roncero, 2018; French translation: Maïano et al., 2022), a 17-item scale that assesses symptoms of orthorexia nervosa (8 items; sample item: "My concern with healthy eating takes up a lot of my time") and "healthy" orthorexia (9 items; sample item: "I spend a lot of time buying, planning, and or/ preparing food so my diet will be as healthy as possible"). This instrument is based on Barrada and Roncero's (2018) model of orthorexia, which distinguishes between an extreme preoccupation with eating healthily (i.e., orthorexia nervosa) and a healthy interest in diet, self-assessed healthy behaviours with regard to diet, and eating healthily as part of one's core identity (i.e., non-psychopathological or "healthy" orthorexia"). Participants rated items on a 4-point scale ranging from 0 (*completely disagree*) to 3 (*completely agree*), with higher scores reflective of greater orthorexic symptomatology. In French-speaking Canadian adults (Maïano et al., 2022), the TOS has been shown to have a 2-factor structure and adequate psychometric properties. In the present study, the composite reliability coefficient (McDonald's ω) of scores on the Orthorexia Nervosa and Healthy Orthorexia subscales were .882 and .873, respectively.

2.2.6. Anxiety and depression

The 14-item Hospital Anxiety and Depression Scale (HADS; Zigmond & Snaith, 1983; French translation: Savard et al., 1998) was used to assess symptoms of anxiety (7 items; sample item: "Worrying thoughts go through my mind") and depression (7 items; sample item: "I feel as if I am slowed down"). Items were rated on a 4-point scale, with total scores for each subscale ranging from 0 to 21. In French-speaking Canadian adults, the HADS has been shown to have good sensitivity to determine the presence of anxiety or depressive symptoms, respectively (Savard et al., 1998). In the present study, the composite reliability coefficients (McDonald's ω) on the Anxiety and Depression subscales were .846 and .818, respectively.

2.2.7. Life satisfaction

Participants completed the 5-item Satisfaction With Life Scale (SWLS; Diener et al., 1985; French translation: Blais et al., 1989) to assess an aspect of subjective well-being (sample item: "So far I have gotten the important things I want in my life"). Its items are rated on a 7-point scale (1 = *strongly disagree*, 7 = *strongly agree*), with higher scores reflecting greater life satisfaction. In French-speaking Canadian adults (Swami, Stieger et al., 2025), the French translation of the SWLS has been shown to have a unidimensional factor structure. In the present study, the composite reliability coefficient (McDonald's ω) for SWLS scores was .917.

2.2.8. Self-compassion

We asked participants to complete the 11-item Short Form of the Self-Compassion Scale (SCS-SF; Raes et al., 2011; French translation: Swami, Tran et al., 2025) that assesses multidimensional facets of self-compassion. Items were rated on a 5-point scale, ranging from 1 (*almost never*) to 5 (*almost always*). Recent work, including in French-speaking Canadian adults, has suggested that the SCS-SF is best conceptualised as consisting of two discrete subscales assessing compassionate (5 items; sample item: "When something painful happens I try to take a balanced view of the situation") and uncompassionate self-responding (6 items; sample item: "When I fail at something that's important to me, I tend to feel alone in my failure") (Swami, Tran et al., 2025). In the present study, only the compassionate self-responding subscale was used; the composite reliability coefficient (McDonald's ω) for scores on this subscale was .806.

2.2.9. Self-esteem

Participants were asked to complete the Single-Item Self-Esteem Scale (SISE; Robins et al., 2001; French translation: Swami et al., 2020b). They rated the item "I have high self-esteem" on a 7-point scale (1 = *not very true of me* to 7 = *very true of me*), with higher scores reflecting higher self-esteem. The SISE has been reported to have adequate evidence of construct validity in adults from diverse various languages and culture, including French-speaking Canadian adults (Swami et al., 2020b).

2.3. Procedure

Data collection was authorised by the research ethics committee of the first and last authors' university (#2025–3656) and took place between November 2024 and April 2025. The present study was conducted following the principles of the Declaration of Helsinki. Participants were primarily recruited in the Canadian province of Quebec via a panel from Léger Opinion and received CAD$1.80 in remuneration (95% of the total sample). Briefly, Léger Opinion is an online survey panel in Canada, operated by a market research firm that recruits individuals from the community. A further 5% of the total sample were recruited via advertising placed on social media, a local newspaper, and the websites of private clinics; these participants did not receive any remuneration. All participants provided digital informed consent before completing an anonymous survey, hosted on LimeSurvey, in which the order of presentation of the instruments described above was pre-determined. All participants took part on a voluntary basis.

2.4. Analytic strategy

To facilitate an EFA-to-CFA strategy, the total sample was randomly split into two separate split-halves of 385 participants each using a computer-generated random seed. Independent *t*-tests and chi-square tests showed no significant differences between the two split-half subsamples as function of age, $t(768) = 0.95$, $p = .341$, $d = 0.07$; BMI, $t(755) = 0.84$, $p = .400$, $d = 0.07$; gender identities, $\chi^2(5) = 4.27$, $p = .512$; and educational qualifications, $\chi^2(9) = 4.74$, $p = .856$. These results suggest that the process of randomly splitting the sample was successful at minimising subsample differences in these key demographics.

2.4.1. First split-half subsample

All analyses were performed using Mplus 8.11 (Muthén & Muthén, 2024). As per Hair et al. (2009), data factorability was assessed using the Kaiser-Meyer-Olkin (KMO) measure of sampling adequacy (which should ideally be ≥.80) and Bartlett's test of sphericity (which should be significant). Three EFAs were performed using the maximum likelihood robust (MLR) estimator (missing data at the FAS item level ranged from 0.26% to 2.86%, M = 1.71%), an oblique geomin rotation, and an epsilon value of .5 (Marsh et al., 2009, 2014). Factor retention was based on Horn's (1965) parallel analysis test and the following fit indices (Hu & Bentler, 1999; Marsh et al., 2005): the Comparative Fit Index (CFI; .90 or .95 indicative of acceptable or good fit to the data, respectively), the Tucker Lewis Index (TLI; .90 or .95 indicative of acceptable or good fit to the data, respectively), and the root mean square error of approximation (RMSEA; .08 or .06 indicative of acceptable or good fit to the data, respectively). Finally, composite reliability of FAS latent factor(s) was based on the McDonald's ω.

2.4.2. Second split-half subsample

Analyses were performed using Mplus 8.11 (Muthén & Muthén, 2024). A CFA was performed using the MLR estimator (missing data at the FAS item level ranged from 0% to 3.90%, M = 1.26%). The number of latent factors to be estimated in the CFA model was based on the best EFA solution retained with the first split-half subsample. Model fit was assessed using the fit indices described above (first split-half subsample).

2.4.3. First and second split-half subsamples

Using Mplus 8.11 (Muthén & Muthén, 2024) MLR estimator, we tested whether the optimal CFA model was invariant across both split-half subsamples. The following sequence was used (Millsap, 2011): (i) configural invariance; and invariance of (ii) loadings; (iii) intercepts; (iv) uniqueness; (v) latent variance; and (vi) latent mean factor. These models (i.e., the subsequent model relative to the previous one) were compared based on changes (Δ) in CFI, TLI, and RMSEA. A sequence was considered as non-invariant when ΔCFI and/or ΔTLI decreases were higher than .01, and/or the increase in ΔRMSEA was higher than .015, (Chen, 2007; Cheung & Rensvold, 2002).

A hybrid MIMIC multiple-group model (Maïano et al., 2023; Morin et al., 2018) was used to examine: (a) presence of DIF and latent mean differences as function of predictors (i.e., participant age, BMI, and gender identity); and (b) the invariance of these predictions across subsamples. The most invariant model identified during the measurement invariance test as function of subsamples was used to develop the hybrid MIMIC model. The following sequence were used to estimate the hybrid MIMIC models (Morin, 2023; Swami et al., 2023): (a) null effects; (b) saturated; and (c) factors only. Prior to the analyses and to facilitate interpretation, age and BMI were standardised. First, these models were examined with all predictions freely estimated across subsamples. Next, these predictions were constrained to be equal across subsamples in the most appropriate Hybrid MIMIC model (Maïano et al., 2023; Morin et al., 2018).

First, it was considered that there is a meaningful association between FAS item responses and predictors when the increase of ΔCFI and/or ΔTLI was $\geq$.01 and/or the decrease of ΔRMSEA was $\geq$.015 between the null effects model and the saturated and factors only models. Second, there is DIF when there is an improvement of goodness of fit statistics for the saturated model relative to the factor only model. Third, there is invariance of the association between FAS item responses and predictors when a lack of substantial improvement is observed between the most appropriate hybrid MIMIC model and an alternative version of this model constraining the examined predictions to be equal across subsamples.

Convergent and concurrent validity of the FAS was examined in the total sample using a structural equation model (SEM) in which the FAS latent factor(s) from the best CFA solution was correlated with the observed scores of measures of positive body image (i.e., body appreciation), eating patterns (i.e., dieting behaviour and orthorexia), psychological well-being (symptoms of anxiety and depression, life satisfaction, self-esteem) and self-compassion. In this SEM, observed scores were correlated altogether.

3. Results

3.1. First split-half subsample

Both Bartlett's test of sphericity, $\chi^2(21) = 1567.25$, $p < .001$, and the Kaiser-Meyer-Olkin measure of sampling adequacy (.91), indicated that the FAS items had adequate common variance for factor analysis with the data from the first split-half subsample. Table 2 presents the results from the EFA models. All factors solution had acceptable-to-good fit to the data (Models 1–1 to 1–3), and results from the parallel analysis test provided support for a unidimensional solution (see Figure S1 in the Supplementary Materials for a visual representation). While results supported extraction of a unidimensional model of the FAS, we also considered fit of 2- and 3-factor models to rule these out as viable alternatives in these analyses. Inspection of the 2-factor solution showed that the first factor comprised six of the seven FAS items and that the single item in the secondary factor provided an improper standardised factor loading (> 1.00). Finally, inspection of the 3-factor solution showed that the first factor comprised two items, the second factor three items, and the third factor two items. Therefore, based on several indicators (i.e., fit indices, analysis of the three factor solutions analyses,

Table 2
Goodness-of-fit statistics for the FAS.

Models§	Samples	N°	Description	$R\chi^2$	df	CFI	TLI	RMSEA	RMSEA 90 % CI		CM	$\Delta R\chi^2$	df	p	ΔCFI	ΔTLI	ΔRMSEA
									LB	UB							
EFA	First-split-half	1–1	1 factor	42.650*	14	.968	.952	.073	.049	.098	-	-	-	-	-	-	-
		1–2	2 factors	26.501*	8	.979	.945	.078	.046	.111	-	-	-	-	-	-	-
		1–3	3 factors	9.289*	3	.993	.950	.074	.023	.130	-	-	-	-	-	-	-
CFA	Second-split-half	2–1	1-factor	20.655	14	.991	.987	.035	.000	.065	-	-	-	-	-	-	-
Measurement invariance across samples	First and second split-half	3–1	Configural invariance	63.015*	28	.979	.968	.057	.038	.076	-	-	-	-	-	-	-
		3–2	Loadings invariance	69.964*	34	.978	.973	.052	.035	.070	3–1	4.68	6	.59	-.001	+.005	-.005
		3–3	Intercepts invariance	82.303*	40	.974	.973	.052	.036	.068	3–2	12.34	6	.05	-.004	.000	.000
		3–4	Uniqueness invariance	96.067*	47	.970	.973	.052	.037	.067	3–3	13.83	7	.05	-.004	.000	.000
		3–5	Variance invariance	96.517*	48	.970	.974	.051	.036	.066	3–4	.31	1	.58	.000	+.001	-.001
		3–6	Latent mean invariance	98.581*	49	.970	.974	.051	.036	.066	3–5	2.10	1	.15	.000	.000	.000
DIF: Age, body mass-index and gender	First and second split-half	4–1	Null effects	239.002*	91	.931	.937	.066	.056	.076	-	-	-	-	-	-	-
		4–2	Saturated	99.450*	49	.977	.960	.052	.037	.067	4–1	157.36	42	< .001	+.046	+.023	-.014
		4–3	Factors only	172.373*	85	.960	.960	.052	.041	.064	4–1	87.90	6	< .001	+.029	+.023	-.014
		4–4	Partial factors only	149.215*	82	.969	.968	.047	.035	.059	4–2	43.63	33	.10	-.008	+.008	-.005
		4–5	Partial factors only (invariant across samples)	152.609*	85	.969	.969	.046	.034	.058	4–4	2.59	3	.46	.000	+.001	-.001

Note. FAS = Functionality Appreciation Scale; $R\chi^2$ = robust maximum likelihood chi-square; *df* = degrees of freedom; CFI = comparative fit index; TLI = Tucker-Lewis index; RMSEA = root mean square error of approximation; 90% CI = 90% confidence interval of the RMSEA; LB = lower bound; UB = upper bound; Δ = change from the previous model; $\Delta R\chi^2$ = robust chi-square difference tests (calculated from loglikelihoods for greater precision); EFA = exploratory factor analyses; CFA = confirmatory factor analysis; DIF = differential item functioning. * $p \leq .05$

and the results of the parallel analysis test), the unidimensional model with all seven items was retained for subsequent analyses. The standardised parameters and the composite reliability coefficient for this model are reported in Table 3. Results showed that factor loading values were high (λ =.690–.865; M_{λ} =.776) and that the composite reliability coefficient of the FAS latent factor was excellent (ω =.914).

3.2. Second Split-half subsample

Table 2 presents the goodness-of-fit statistics from the unidimensional model. Results showed an acceptable fit to the data and, as illustrated in Table 3, factor loading values were high (λ =.693–.802; M_{λ} =.768). The composite reliability coefficient of the FAS latent factor was excellent (ω =.910).

3.3. First and second split-half subsamples

3.3.1. Subsample invariance

Table 2 also presents the goodness-of-fit statistics from tests of measurement invariance of the unidimensional CFA-based model as function of split-half subsamples (Models 3–1 to 3–6). Results provided support for the complete measurement invariance (i.e., configural, loadings, intercepts, uniqueness, variance and latent mean) of this model.

3.3.2. Differential item functioning across both subsamples

Table 2 provides the goodness-of-fit statistics of the hybrid MIMIC models. The first model starts with the latent mean invariance model (Model 3–6), which was the most invariant model across split-half subsamples. First, results revealed that the FAS responses were associated with participant age, BMI, and gender identity. Second, the improvement in model fit of the saturated model relative to the factors only model (ΔCFI = +.017, ΔTLI =.000, ΔRMSEA =.000) indicated DIF as a function of predictors. Modification indices of the factors-only model were inspected and revealed that DIF could be attributed to one path from age to one FAS item (i.e., Item #7 - "I respect my body for the functions it performs" - from the second-split half subsample) and two paths from BMI to FAS items (i.e., Item #3 - "I appreciate that my body allows me to communicate and interact with others" - from the first-split half subsample; Item #1 - "I appreciate my body for what it is capable of doing" - from the second-split half subsample). Therefore, these constraints were relaxed, and the resulting partial factors-only model did not show a substantial decrease of fit indices relative to the saturated model. Finally, the last model (Model 4–5, invariance of the partial factors-only model) confirmed the invariance across subsamples of the associations between the FAS latent factor and age, BMI, and gender identity.

Table 4 provides the results from the partial factors-only model invariance tests across subsamples. Results showed a lack of associations between gender identity and the FAS latent factor. However, a significant association was found between the FAS latent factor and age and BMI, respectively. More specifically, the results showed that older adults tended to present significantly higher levels of functionality appreciation relative to younger participants. In addition, participants with higher BMIs tended to present significantly lower levels of functionality appreciation relative to participants with lower BMIs.

Table 3
Standardised parameters estimates from the exploratory and confirmatory factor analytic representation of the FAS in the first and second-split half subsamples.

	First split-half subsample		Second split-half subsample	
Items	λ	δ	λ	Δ
1	.756	.429	.792	.372
2	.797	.365	.790	.376
3	.716	.487	.758	.425
4	.690	.524	.693	.520
5	.764	.417	.757	.427
6	.865	.251	.785	.383
7	.843	.290	.802	.357
ω	.914		.910	

Note. FAS = Functionality Appreciation Scale; λ = factor loadings; δ = uniquenesses; ω = McDonald's omega.

Table 4
Relationships between the FAS latent factor and the predictors.

		Subsamples-specific standardised coefficients	
	b(SE)	β (First split-half subsample)	β (Second split-half subsample)
Age	.084(.036)*	.080*	.079*
Body mass index	-.353(.045)***	-.332***	-.332**
Gender identity	.055(.077)	.026	.026

Note. FAS = Functionality Appreciation Scale; b = unstandardised regression coefficient taken from the partial Factors only invariant across samples (Model 4–5); SE = standard error of the coefficient; β = subsample-specific standardised regression coefficient (although some of the relations are invariant across subsamples, the standardized coefficients may still show some variation as a function of within-subsamples estimates of variability). Because age and body mass index were standardised prior to these analyses and that the FAS factor is estimated based on a model of latent variance invariant in which all latent factors have a *SD* of 1, all unstandardised coefficients can be directly interpreted is *SD* units. * $p \leq .05$; ** $p \leq .01$; *** $p \leq .001$.

3.3.3. Convergent and concurrent validity

Goodness-of-fit statistics from the SEM examining the convergent and concurrent validity of the unidimensional model of the FAS provided a good level of fit to the data: $\chi^2(68)$ = 192.503, CFI = .975, TLI = .956, RMSEA = .049 (90% CI =.041; .057). Results provided in Table 5 revealed significant and positive relationships between functionality appreciation and body appreciation, self-compassion, life acceptation, healthy orthorexia, and self-esteem. On the other hand, significant and negative relationships were found between functionality appreciation and orthorexia nervosa, anxiety, depression, and dieting, respectively.

4. Discussion

In the present study, our objective was to examine the psychometric properties of a novel French translation of the FAS in an online sample of French-Canadian adults. Overall, our results corroborate other test adaptation studies (see Table 1) in showing that the FAS retains robust psychometric properties in our sample. More specifically, in terms of factorial validity, our results supported a unidimensional model of the FAS with all seven items retained. This model was fully invariant across both split-half subsamples in the present study. In addition, across subsamples, we found some evidence of DIF as a function of participant age and BMI, which could be attributed to three paths across the two

Table 5
Construct validity of the FAS in the overall sample.

	FAS
Body appreciation	.704**
Self-compassion	.435**
Life satisfaction	.545**
Healthy orthorexia	.256**
Orthorexia nervosa	-.124*
Anxiety	-.345**
Depression	-.420**
Self-esteem	.457**
Dieting	-.332**

Note. FAS = Functionality Appreciation Scale. * $p \leq .01$; ** $p \leq .001$.

subsamples. Based on a partial factors-only model, we found that that greater functionality appreciation was associated with older age and lower BMIs. Finally, we also found strong evidence of convergent and concurrent validity for the French FAS, with all associations with additional constructs in the expected direction.

In terms of factorial validity, our EFA-based results from the first split-half subsample indicated that a unidimensional model of the FAS should be extracted, with all seven items loading well onto the latent factor. This unidimensional model also showed good fit to the data based on our CFA results in the second split-half subsample. Additionally, this model was fully invariant across both subsamples, supporting the robustness of the unidimensional model of the FAS. Overall, then, the present results suggest that the FAS taps a unidimensional latent functionality appreciation construct in our French-Canadian sample. This is consistent with the findings of the parent study reporting on the development of the FAS (Alleva et al., 2017), as well as all previous test adaptation studies (see Table 1).

In addition, across subsamples, we examined DIF in responses to the FAS based on participant age and BMI. Our results showed some minor DIF, with a total of three paths involved – one in age and two in BMI – across the two subsamples. Relaxing the constraints of these paths in a factors-only model allowed us to achieve a final model that was invariant across split-half subsamples. In this model, we found a lack of DIF across gender identity and that it was not significantly associated with functionality appreciation. In broad outline, this finding is consistent with previous reports that FAS factor structure and scores are invariant across women and men (see Table 1), although it should be noted that earlier studies have assessed gender invariance in isolation of other demographic factors. Our results are also consistent with the meta-analytic finding that there are no significant gender differences in functionality appreciation (Linardon et al., 2023). While gender differences in functionality appreciation may be larger in some national contexts (e.g., Mebarak et al., 2023), it may be that in the French-Canadian context, there are insufficient motivators for gendered differences (e.g., in terms of appearance ideals geared toward physical prowess for men).

In terms of age, our results showed that older participants tended to have higher functionality appreciation that younger participants. Although Linardon et al. (2023) reported no significant association between functionality appreciation age in their meta-analysis, several recent studies have reported that older participants have higher functionality appreciation than younger participants (e.g., He et al., 2023; Mebarak et al., 2024). One likely explanation for this finding is that older adults have greater opportunities to deprioritise an aesthetic view of the body and experience lower sociocultural pressure to focus on the body's appearance, though there may be gendered differences in how this manifests in some national contexts (Bennett et al., 2020; Jankowski et al., 2016). Instead, older adults may have greater time and space to value the body's functions and capabilities (Alleva, 2025).

In fact, the finding that functionality appreciation was positively associated with age is particularly noteworthy given that limitations in body functionality often *increase* with age (Maresova et al., 2023). This finding supports Alleva et al. (2017) assertion that functionality appreciation is distinct from body functionality and challenges a lay assumption that those with functional impairments cannot holistically appreciate their body functionality (for additional evidence, see Bailey et al., 2015; Vinoski Thomas et al., 2019). Additionally, we found that participants with lower BMIs reported greater levels of functionality appreciation, which is consistent with previous work (Linardon et al., 2023) and could reflect the fact that larger-bodied individuals experience greater weight-based stigma and discrimination (e.g., Pearl, 2018), leaving them with fewer opportunities to appreciate the functions of their bodies.

Finally, the present results also showed that our French translation of the FAS has strong convergent and concurrent validity in our sample. More specifically, convergent validity was supported insofar as greater functionality appreciation was significantly associated higher body appreciation and lower maladaptive eating patterns (dieting behaviour and orthorexia nervosa). This is potentially important from a practical point of view because it suggests that functionality appreciation, like with other indices of positive body image, may be associated with fewer negative eating habits (Linardon et al., 2023). This, in turn, may mean that interventions can be leveraged to promote functionality appreciation as one route to mitigate against maladaptive eating practices. Indeed, recent qualitative work with individuals with lived experience of an eating disorder or body image distress has shown that they view the promotion of functionality appreciation as useful, complementing existing treatment approaches, and important for early intervention (Mulgrew et al., 2024).

One notable point, however, was the strong association between body appreciation and functionality appreciation (see Table 5), which suggests a high degree of nomological overlap (approximately 50%) between these constructs. Similar findings have been reported elsewhere (e.g., Cerea et al., 2021) and may require some attention from scholars. For example, it may be that, in some national or linguistic contexts, there is a higher degree of construct overlap between body appreciation and functionality appreciation, which may have both theoretical (i.e., how these constructs are defined and delineated) and practical implications (e.g., minimising construct redundancy in survey designs). Yet, in the positive body image literature (see, for example, Alleva & Tylka, 2021; Tylka & Wood-Barcalow, 2015a; Wood-Barcalow et al., 2010), functionality appreciation is conceptualised as a more specific type of body appreciation, so the strong correlation between these constructs should be expected.

Our results also support the concurrent validity of the FAS in the present sample. Specifically, we found that greater functionality appreciation was significantly associated with higher psychological well-being, as indexed by fewer symptoms of anxiety and depression, as well as higher life satisfaction and self-esteem. This is consistent with the findings of a meta-analysis showing that functionality appreciation is significantly associated with indices of well-being ($r = .33$; Linardon et al., 2023). Finally, the present results also showed that greater functionality appreciation was significantly associated with higher self-compassion, which again is consistent with previously reported effects ($r = .37$; Linardon et al., 2023). Overall, then, the results of the present study are both consistent with previous work and support the psychometric properties of the French FAS in French-Canadian adults.

Nevertheless, there are several issues that limit the generalisability of our findings. First, in terms of our study design, we did not assess the temporal stability or test-retest reliability of our data, which should be rectified in the future. Second, our survey did not include any instruments that would have allowed us to assess other forms of validity, such as discriminant validity. Third, also related to our study design, given that the FAS was originally developed for use with adults, we did not assess its psychometric properties in younger age groups. Doing so may be useful in the future (see He et al., 2023), particularly if researchers wish to know to what extent the FAS can be used confidently in younger age groups. If doing so, the extent to which the FAS meaningfully captures all relevant facets of functionality appreciation in younger age groups should first be determined (see Craddock et al., 2025). Fourth, it should be noted that the use of the BMI as index of body size has been problematised (Bailey et al., 2025) and, in view of such critiques, we suggest that all analyses that included BMI should be treated with caution. In future work, it may be expedient to substitute BMI with other relevant measures, such as self-descriptions of body size or clothing size.

In terms of our sample, it should be noted that the recruitment method – which utilised an online panel and online community sampling – means that we were unlikely to have recruited a representative sample of Québécois participants. Our sampling strategy may have also inadvertently introduced sampling biases (e.g., some participants being more likely to respond to an advertisement about a survey on body image), which again limits the generalisability of our findings. We also did not gather additional sociodemographic data about our sample, such as information about urbanicity, socioeconomic status and food insecurity, and religion or religiosity, which may have important intersectional impacts on functionality appreciation. Related to sampling is the fact that our sample was limited to French Canadians. In the future, it may be useful to examine the extent to which the FAS is invariant across minoritised communities in Quebec (e.g., First Nation peoples, as well as across Francophone and Anglophone Canadians), as well as with participants with diverse social identities (i.e., racialised minorities and individuals with a disability/chronic health condition, and neurodivergence). Indeed, these populations may experience functionality appreciation differently than the general population. Finally, a small minority of participants reported having (currently or in the past) an eating disorder diagnosis or were pregnant at the time of study, and we do not know how these characteristics may have influenced our results.

These limitations notwithstanding, our findings indicate that our novel French translation of the FAS retains it psychometric properties in our sample of French-Canadian adults. This is important, not only because it adds to the growing list of populations for which the FAS has been validated, but also because it adds to the evidence that the FAS is measuring a common latent construct of functionality appreciation in various national, cultural and/or linguistic groups. Given that similar findings have been reported in diverse national contexts (see Table 1), an important next step for researchers will be to assess the degree to which the FAS is invariant across national and linguistic groups (see Mebarak et al., 2024) – as it has been conducted for other measures of positive body image, (e.g., the BAS-2; Swami, Tran et al., 2023). Doing so will help researchers better understand the ways in which functionality appreciation may (or may not) differ across geographic contexts, as well as the ways in which cultural values and factors may shape functionality appreciation.

CRediT authorship contribution statement

Christophe Maïano: Conceptualization, Methodology, Validation, Formal analysis, Data curation, Resources, Writing – original draft, Writing – review & editing, Software, Supervision, Visualization, Project administration, Investigation, Funding acquisition. **Viren Swami:** Conceptualization, Methodology, Validation, Writing – original draft, Writing – review & editing, Supervision. **Tracy L. Tylka:** Conceptualization, Methodology, Validation, Writing – review & editing. **Annie Aimé:** Conceptualization, Methodology, Validation, Writing – review & editing.

Ethical statement

This study was approved by the research ethics committee of the Université du Québec en Outaouais (#2025–3656) and conducted in accordance with the Declaration of Helsinki. Participants provided digital consent prior to responding to the online survey.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper. Given her role as Editor-in-Chief of Body Image, Tracy L. Tylka had no involvement in the peer review of this article and had no access to information regarding its peer review. Full responsibility for the editorial process for this article was delegated to another journal editor.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.bodyim.2026.102044.

Data availability

The authors do not have permission to share data.

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